



RPC Implementation in Cloud Environment

ASSIGNMENT-1



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KAVIPRAKASH J
2023115104

Aim

To design and implement a **Remote Procedure Call (RPC) based distributed application** using **Python**, where the **server is hosted in a cloud environment (Microsoft Azure VM)** and the **client runs on a local machine**, enabling remote procedure invocation over a network.

Software & Tools Used

Component	Description
Programming Language	Python 3
RPC Mechanism	HTTP-based RPC
Cloud Platform	Microsoft Azure
Server OS	Ubuntu Linux (Azure VM)
Client OS	Windows 10
Libraries Used	http.server, json, requests

System Architecture

- The **RPC Server** is deployed on an **Azure Virtual Machine**
- The **Client** runs on a **local Windows system**

- Client sends requests using HTTP POST
- Server processes the request and returns results in JSON format

Client (Windows) → Azure VM (RPC Server)

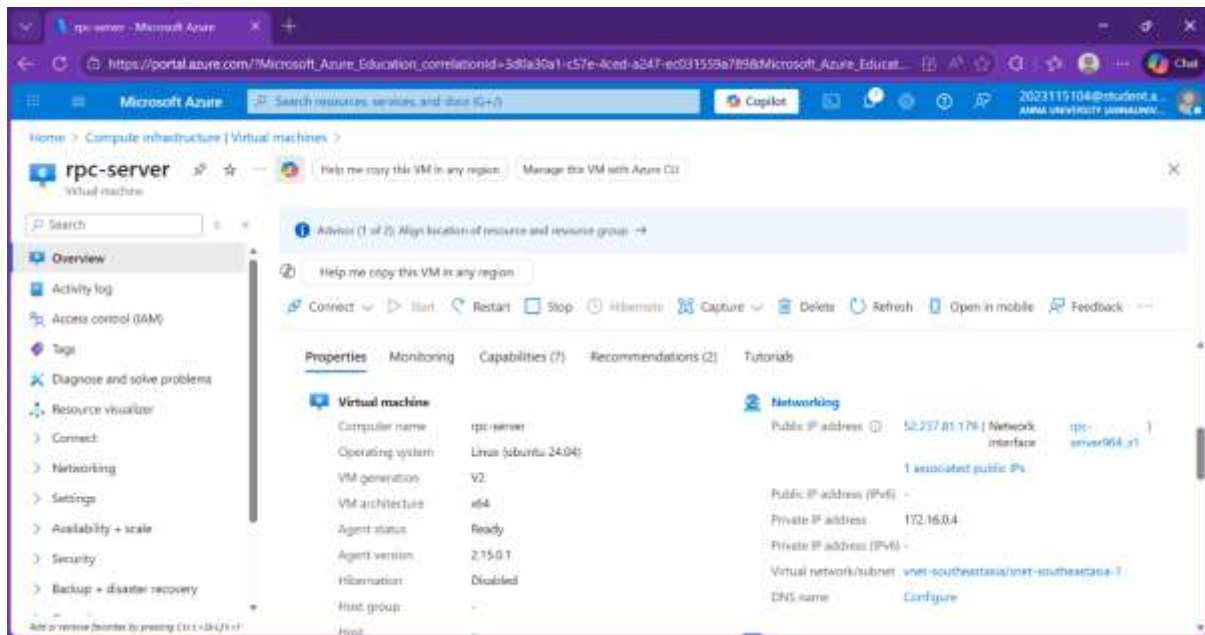
Remote Procedures Implemented

Procedure Name	Description
word_count	Counts number of words in a given text
reverse_text	Reverses the given string
is_palindrome	Checks whether the string is a palindrome

Execution Steps

Step 1: Start Azure VM

- Login to Azure Portal
- Start Ubuntu Virtual Machine
- Enable inbound rule for port **8000**



Step 2: Run RPC Server on Azure VM

`python3 rpc_server.py`

Output:

RPC Server running on Azure VM at port 8000

```
azureuser@rpc-server: ~$ cat /etc/os-release
System information as of Thu Jan 29 17:39:32 UTC 2026
System load: 0.08      Processes: 135
Usage of /: 5.6% of 28.02GB   Users logged in: 0
Memory usage: 3%      IPv4 address for eth0: 172.16.0.4
Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

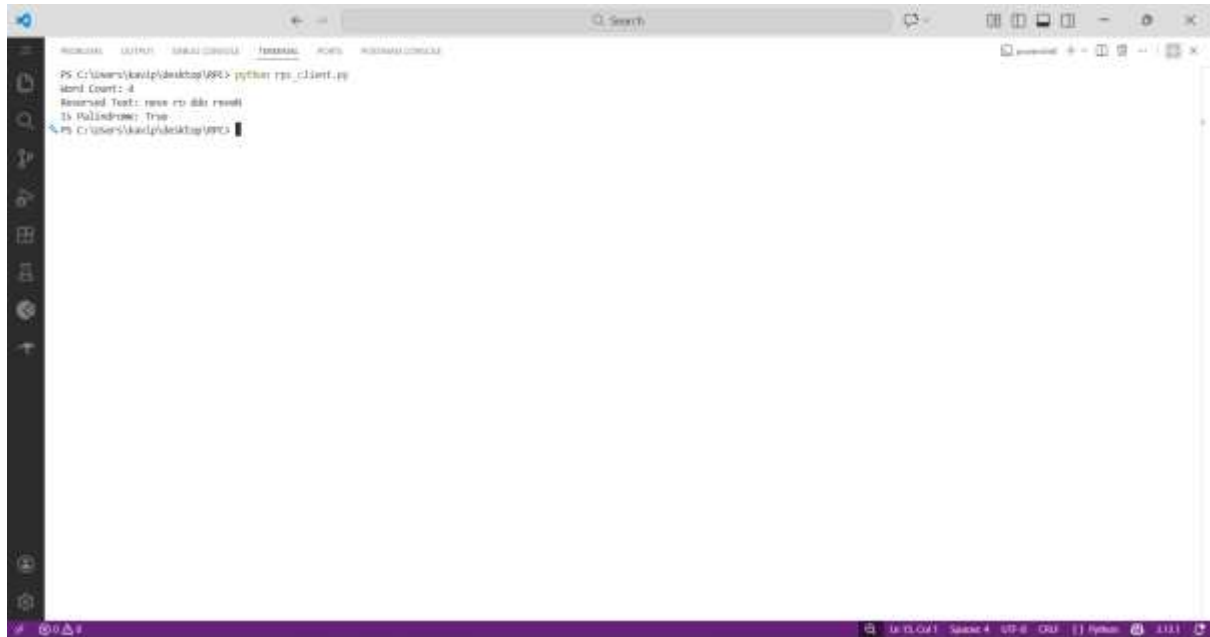
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

azureuser@rpc-server:~$ nano server.py
azureuser@rpc-server:~$ python3 server.py
RPC Server running on Azure VM at port 8000
```

Step 3: Run Client on Local Machine

`python rpc_client.py`



The screenshot shows a Windows command prompt window with the following text:

```
PS C:\Users\kavip\Desktop\RPC> python rpc_client.py
Word Count: 4
Reversed Text: neve ro ddo reveN
Is Palindrome: True
PS C:\Users\kavip\Desktop\RPC>
```

The window title bar indicates it is a "powershell" window. The taskbar at the bottom shows the system clock as 10:15, the date as Oct 1, and the system tray includes icons for Taskbar, UTF-8, CPU, Python, and a version number 3.12.1.

Output

Word Count: 4

Reversed Text: neve ro ddo reveN

Is Palindrome: True

Error Handling

- Invalid RPC endpoints return **404 error**
 - JSON parsing errors handled by server
 - Network connectivity verified using Azure NSG rules
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Result

Thus, a **Remote Procedure Call (RPC) based distributed application** was successfully implemented and executed in a **cloud environment using Microsoft Azure**, allowing a remote client to invoke procedures hosted on the server and receive correct results.

Conclusion

The experiment demonstrates how RPC enables transparent communication between distributed systems. Hosting the server in the cloud allows scalability and remote access, making RPC suitable for real-world distributed applications.
